VITYARTHI PROJECT

AI STUDY ASSISTANT

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BRING YOUR OWN PROJECT (BYOP)

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INTRODUCTION

Students often struggle with organizing their studies, tracking exams, and maintaining academic performance. This project presents an AI Study Assistant that helps students plan their studies, calculate CGPA, manage exams, and receive study suggestions. The system is built as a CLI-based modular application using Python.

PROBLEM STATEMENT

Students face several challenges such as:

- Difficulty planning study schedules
- Poor time management
- Tracking upcoming exams
- Maintaining academic performance
- Lack of motivation while studying

This project aims to solve these problems by creating an AI-based study assistant that helps students organize and improve their academic workflow.

MOTIVATION

The motivation behind this project was to create a simple yet effective tool that helps students improve productivity and academic performance. Many students rely on manual planning, which is inefficient and inconsistent.

By developing an AI Smart Study Assistant, students can automate study planning, track exams, and receive intelligent recommendations.

WHAT I LEARNED

- Understood modular programming and organizing code into multiple files
- Building command-line interface (CLI) applications
- Implementing rule-based AI logic for decision making
- Handling user input and input validation
- Designing user-friendly program flow
- Structuring a real-world software project
- Using GitHub and Git for version control and managing commits

IMPLEMENTATION & FEATURES

1. **main.py:** This file controls the overall application flow. It displays the menu, accepts user input, and calls the appropriate functions from other modules.

2. **planner.py:** This file generates study plans based on subject difficulty. Users input subjects and their difficulty levels, and the system suggests study methods and recommendations.

3. **cgpa.py:** This file calculates CGPA based on user-entered grades.

4. **emotion.py:** This file detects user mood based on input and suggests appropriate study strategies. It uses rule-based logic to provide recommendations.

5. **tips.py:** This file provides random study tips to improve productivity and study habits.

6. **exam.py:** This module allows users to add exams and track upcoming exams. It calculates the remaining days and displays upcoming exams.

TESTING APPROACH

Study Planner Testing

- Tested with different numbers of subjects
- Tested various difficulty levels
- Verified correct study recommendations

CGPA Calculator Testing

- Tested valid grades between 0–10
- Tested invalid inputs (greater than 10 and negative values)
- Verified correct CGPA calculation

Emotion Module Testing

- Tested different mood inputs such as tired, stressed, and motivated
- Verified appropriate study suggestions

Study Tips Testing

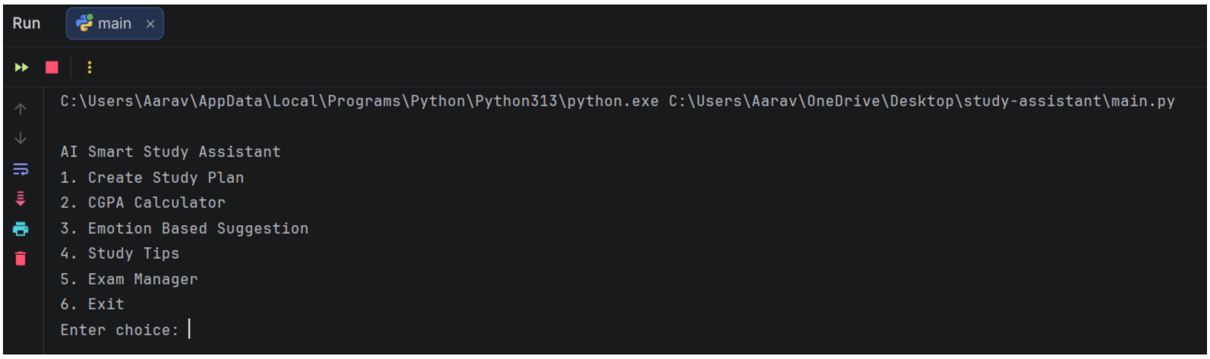
- Tested multiple runs to ensure random tips were displayed

Exam Manager Testing

- Tested adding exams
- Tested viewing exams
- Verified correct calculation of days remaining

SCREENSHOTS & WORKING

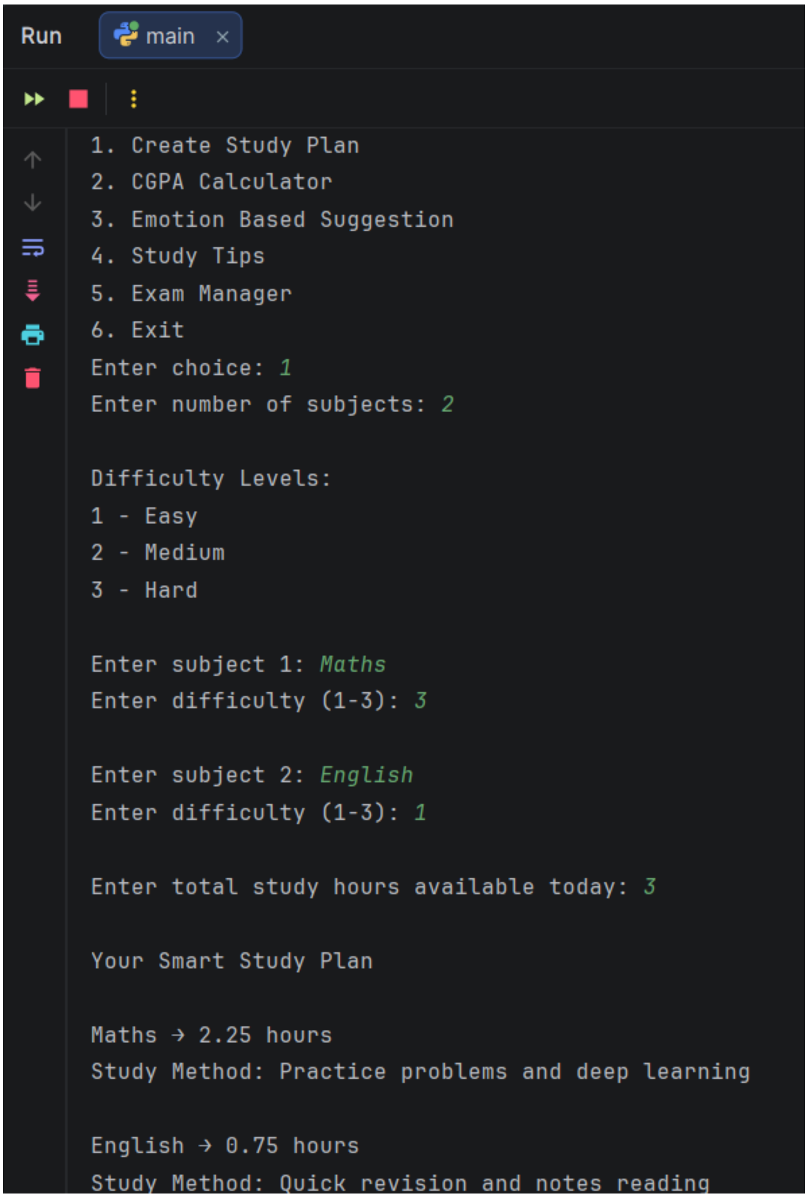
1. Main Menu:



```
C:\Users\Aarav\AppData\Local\Programs\Python\Python313\python.exe C:\Users\Aarav\OneDrive\Desktop\study-assistant\main.py

AI Smart Study Assistant
1. Create Study Plan
2. CGPA Calculator
3. Emotion Based Suggestion
4. Study Tips
5. Exam Manager
6. Exit
Enter choice:
```

2. Study Planner:



```
1. Create Study Plan
2. CGPA Calculator
3. Emotion Based Suggestion
4. Study Tips
5. Exam Manager
6. Exit
Enter choice: 1
Enter number of subjects: 2

Difficulty Levels:
1 - Easy
2 - Medium
3 - Hard

Enter subject 1: Maths
Enter difficulty (1-3): 3

Enter subject 2: English
Enter difficulty (1-3): 1

Enter total study hours available today: 3

Your Smart Study Plan

Maths → 2.25 hours
Study Method: Practice problems and deep learning

English → 0.75 hours
Study Method: Quick revision and notes reading
```

SCREENSHOTS & WORKING

3. CGPA Calculator

```
Run   main ×

C:\Users\Aarav\AppData\Local\Programs\Python\Python313\python.exe C:\Users\Aarav\OneDrive\Desktop\study-assistant\main.py

AI Smart Study Assistant
1. Create Study Plan
2. CGPA Calculator
3. Emotion Based Suggestion
4. Study Tips
5. Exam Manager
6. Exit
Enter choice: 2
Enter number of subjects: 2

Enter grades out of 10
Enter grade for subject 1: 8
Enter grade for subject 2: 8.5

Your CGPA is:  8.25

Press Enter to return to menu...
```

4. Emotion-Based Suggestion

```
Run   main ×

C:\Users\Aarav\AppData\Local\Programs\Python\Python313\python.exe C:\Users\Aarav\OneDrive\Desktop\study-assistant\main.py

AI Smart Study Assistant
1. Create Study Plan
2. CGPA Calculator
3. Emotion Based Suggestion
4. Study Tips
5. Exam Manager
6. Exit
Enter choice: 3
How are you feeling today? happy

Detected Mood: Motivated
Study Suggestion: Start with the hardest subject.

Press Enter to return to menu...
```

SCREENSHOTS & WORKING

5. Exam Manager

```
Run   main ×

C:\Users\Aarav\AppData\Local\Programs\Python\Python313\python.exe C:\Users\Aarav\OneDrive\Desktop\study-assistant\main.py

AI Smart Study Assistant
1. Create Study Plan
2. CGPA Calculator
3. Emotion Based Suggestion
4. Study Tips
5. Exam Manager
6. Exit
Enter choice: 5

Exam Manager
1. Add Exam
2. View Exams
3. Back
Enter choice: 1
Enter subject name: Maths
Days until exam: 5
Exam added successfully

Exam Manager
1. Add Exam
2. View Exams
3. Back
Enter choice: 2

Upcoming Exams:
Maths - 5 days left
```

6. Study Tips

```
Run   main ×

C:\Users\Aarav\AppData\Local\Programs\Python\Python313\python.exe C:\Users\Aarav\OneDrive\Desktop\study-assistant\main.py

AI Smart Study Assistant
1. Create Study Plan
2. CGPA Calculator
3. Emotion Based Suggestion
4. Study Tips
5. Exam Manager
6. Exit
Enter choice: 4

AI Study Tip
Start with the hardest subject first.

Press Enter to return to menu...
```

CHALLENGES FACED

While making this project, I faced several challenges:

- Designing modular architecture
- Implementing user-friendly CLI interface
- Creating meaningful AI-based recommendations
- Handling user input validation

I resolved these issues by placing checks for taking correct input and by referring to various online sources.

KEY DECISIONS MADE

Some key decisions made during development:

- Using CLI for simplicity
- Modular file structure for ease of readability
- Rule-based AI instead of complex ML models
- Simple user-friendly interface

These decisions helped keep the project practical and manageable.

RESULTS

The final system successfully:

- Generates study plans
- Calculates CGPA
- Tracks exams
- Provides study suggestions
- Improves student productivity

The application runs smoothly and provides meaningful assistance to students.

FUTURE ENHANCEMENTS

Possible improvements include:

- Goal CGPA calculator
- Data saving functionality
- Weekly planner
- GUI interface
- More AI-based recommendations

TECHNOLOGIES/TOOLS USED

1. Python3 (Programming language)

2. PyCharm (IDE)

3. Git, GitHub and GitKraken (Version Control)

REFERENCES

1. Python Documentation
2. Course Materials
3. Online Programming Resources